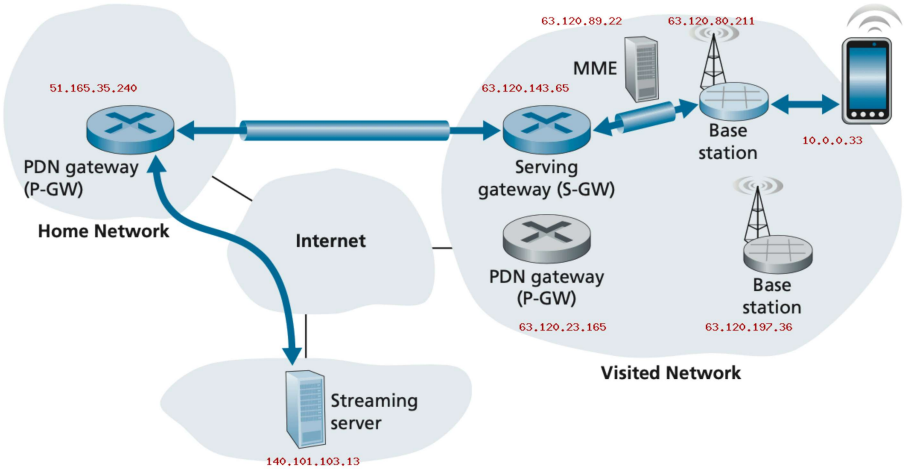
INTERACTIVE END-OF-CHAPTER EXERCISES

4G WIRELESS TUNNELING

In the graphic below, a mobile phone has a TCP connection with the streaming server using wireless 4G. Assume that both the mobile phone and the server use TCP port 11556 and any intermediary nodes that tunnel the datagrams use port 5678.



QUESTION LIST

1. What is the source IP address of the datagram sent from the mobile phone to the base station?

2. What is the destination IP address of the datagram sent from the mobile phone to the base station?

3. What is the transport-layer port number in the datagram sent from the mobile phone to the base station?

4. What transport-layer protocol is indicated in the datagram sent from the mobile phone to the base station?

5. Is there an encapsulated datagram within the datagram sent from the mobile phone to the base station? Answer Yes or No

6. What is the source IP address of the datagram sent from the base station to the serving gateway?

7. What is the destination IP address of the datagram sent from the base station to the serving gateway?

8. What is the transport-layer port number in the datagram sent from the base station to the serving gateway?

9. What transport-layer protocol is indicated in the datagram sent from the base station to the serving gateway?

10. Is there an encapsulated datagram within the datagram sent from the base station to the serving gateway? Answer Yes or No

11. What is the source IP address of the encapsulated datagram?

12. What is the destination IP address of the encapsulated datagram?

13. What is the transport-layer port number in the encapsulated datagram?

14. What transport-layer protocol is indicated in the encapsulated datagram?

15. What is the source IP address of the datagram sent from the serving gateway to the PDN gateway?

16. What is the destination IP address of the datagram sent from the serving gateway to the PDN gateway?

17. What is the transport-layer port number in the datagram sent from the serving gateway to the PDN gateway?

18. What transport-layer protocol is indicated in the datagram sent from the serving gateway to the PDN gateway?

19. Is there an encapsulated datagram within the datagram sent from the serving gateway to the PDN gateway? Answer Yes or No

20. What is the source IP address of the encapsulated datagram?

21. What is the destination IP address of the encapsulated datagram?

22. What is the transport-layer port number in the encapsulated datagram?

23. What transport-layer protocol is indicated in the encapsulated datagram?

24. What is the source IP address of the datagram sent from the PDN gateway to the server?

25. What is the destination IP address of the datagram sent from the PDN gateway to the server?

26. What is the transport-layer port number in the datagram sent from the PDN gateway to the server?

27. What transport-layer protocol is indicated in the datagram sent from the PDN gateway to the server?

28. Is there an encapsulated datagram within the datagram sent from the PDN gateway to the server? Answer Yes or No

29. Is data encrypted between the mobile phone and the base station? Answer Yes or No

30. Does the server know about any of the wireless tunneling? Answer Yes or No

SOLUTION

1. From the mobile phone to the base station, the source address is 10.0.0.33

2. From the mobile phone to the base station, the destination address is 140.101.103.13

3. From the mobile phone to the base station, the port used is 11556

4. From the mobile phone to the base station, the protocol used is TCP

5. Is there an encapsulated datagram? No

6. From the base station to the serving gateway, the source address is 63.120.80.211

7. From the base station to the serving gateway, the destination address is 63.120.143.65

8. From the base station to the serving gateway, the port used is 5678

9. From the base station to the serving gateway, the protocol used is UDP

10. Is there an encapsulated datagram? Yes

11. The source address of the encapsulated datagram is 10.0.0.33

12. The destination address of the encapsulated datagram is 140.101.103.13

13. The port used in the encapsulated datagram is 11556

14. The protocol used in the encapsulated datagram is TCP

15. From the serving gateway to the PDN gateway, the source address is 63.120.143.65

16. From the serving gateway to the PDN gateway, the destination address is 51.165.35.240

17. From the serving gateway to the PDN gateway, the port used is 5678

18. From the serving gateway to the PDN gateway, the protocol used is UDP

19. Is there an encapsulated datagram? Yes

20. The source address of the encapsulated datagram is 10.0.0.33

21. The destination address of the encapsulated datagram is 140.101.103.13

22. The port used by the encapsulated datagram is 11556

23. The protocol used by the encapsulated datagram is TCP

24. From the PDN gateway to the server, the source address is 51.165.35.240

25. From the PDN gateway to the server, the destination address is 140.101.103.13

26. From the PDN gateway to the server, the port used is 11556

27. From the PDN gateway to the server, the protocol used is TCP

28. Is there an encapsulated datagram? No

29. Yes, the base station and mobile phone encrypt data between them

30. No, the server is blissfully unaware of the wireless infrastructure or tunneling

QUESTION 1 OF 30

What is the source IP address of the datagram sent from the mobile phone to the base station?

Answer

CHECK HINT SKIP

We gratefully acknowledge the programming and problem design work of John Broderick (UMass '21), which has really helped to substantially improve this site.

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